

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1 a) Write a short note on "Algorithms" and "Flowcharts". 4 CO1, BL1
- b) Write a program to check if an entered number is prime or not. 3 CO1, BL1
- Q.2 a) Write a program to create an array of "n" elements and find the sum of all the elements in the array. 5 CO2, BL3
- b) Write a program to copy a string with spaces into a string variable "college_name" through user input, reverse the string, and find the length of the string using string functions. 5 CO2, BL3
- Q.3 a) A train is scheduled to travel between two cities A and B. There are 'n' halting stations for the journey such that the train halts at specific stations based on the distance between the stations. The first halt is at a distance of 2km from city A and the second halt is at 3km from the first halt. The remaining halts are at a distance of the sum of the distances of the previous two halts. Write a program to print the distance of each halt one after the other up to 'n' halts to reach city B using the concept of functions in C. 5 CO3, BL4
- b) Write a short note on "Call by Value" and "Call by Reference". 3 CO3, BL1
- c) Enlist all the "Storage Classes" in C. Explain any one storage class in detail. 2 CO3, BL2
- Q.4 a) Write a program to create a structure "time" with its members as "hours", "minutes" and "seconds". Create another structure "Car" having the following members as "Car_No", "Start_Time", and "End_Time". Hint: use structure within structure. 5 CO4, BL4
- b) Create a structure variable for the structure created in "Q. 4a" and get the input for each member from the user and display it. 5 CO4, BL3
- Q.5 a) Write a program to swap values of two variables using pointers in C. 4 CO5, BL2
- b) Discuss about the "*" and "&" operators in C. Show its usage in a C program. 4 CO5, BL2



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Seat No.						
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Batch: 2022-26
Semester: I
Course: Programming in C
Course Code: 0705210106

Batch: 2022-26 (Sem I)
Backlog April/Even 2023 Examination

Date: 23/06/2023

Day: Friday

Maximum Marks: 45

Time: 10:00 - 11:30

Instructions:

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1 a) Differentiate between entry and exit controlled loops. 4 CO1, BL4
b) Design an algorithm to print the factorial of a user-given number. 3 CO1, BL3
- Q.2 a) State the rules for constructing variable names of arrays. 5 CO2, BL2
b) Write a program to print the transpose of a user-given matrix. 5 CO2, BL3
- Q.3 a) Write a program to swap two numbers using call by value and call by reference. (Note: Numbers should be entered by user) 5 CO3, BL3
b) Write a program to print the multiplication table of a number using a function. 3 CO3, BL2
c) Explain with an example the actual and formal parameters in a function. 2 CO3, BL3
- Q.4 a) Design a program to create a structure named student having name, rollno, and section as its member. Create a semester structure within the student structure having subjectname, and marks as its members. Ask the user to enter the details of two students and display their marks. (Note: Apply the concept of nested structure) 5 CO4, BL4
b) Demonstrate the need for an array of structures with an example. 5 CO4, BL3
- Q.5 a) Write a program to create a pointer variable, initialize it, and later update its value. 4 CO5, BL3
b) Write short note on Dynamic Memory Allocation in C. 4 CO5, BL2

Batch: 2022-26 (Sem I)
Backlog April/Even 2023 Examination

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Seat No.						
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(0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

(070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: II

Course: Programming in C

Course Code: 0705210215

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination

Date: 14/07/2023

Day: Friday

Maximum Marks: 45

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

a) Use Ball Pen/ Ink Pen (Blue only) to answer all questions.

b) All questions carry marks as indicated.

Solve the following MCQ, and select the most appropriate 10M CO1, BL [2]
choice.[All are compulsory:1 Marks Each]

1.1 Which of the following declarations is not supported by the C language?

- a. String str="34.56";
- b. char *str;
- c. float str=3e2;
- d. a and c both

1.2 In a programming language, if signed integer is stored in 6 bytes then, what is the range of values stored in the integer variable

- a. $-(2^{31})$ to $(2^{31}) - 1$
- b. $-(2^{48})$ to $(2^{48}) - 1$
- c. $-(2^{47})$ to $(2^{47}) - 1$
- d. 0 to $(2^{48}) - 1$

1.3 What is the output of the following code?

```
char a = 'A', *b = &a; //Line 1
int c = 5, *d = &c; // Line 2
if(sizeof(b) == sizeof(d))
    printf("Same");
else
    printf("Difference");
```

- a. Error in Line 1 and Line 2
- b. Same
- c. Difference
- d. None of the above

1.4 What is the output of the following code?

```
char a = 'A', *b = &a; //Line 1
int c = 5, *d = &c; // Line 2
if(sizeof(*b) == sizeof(*d))
    printf("Same");
else
    printf("Difference");
```


- a. Error in Line 1 and Line 2
- b. Same
- c. Difference
- d. None of the above

1.5 What is the output of the following code?

```
int a = 0, b = 1, c, d;
c = a++ && b--;
printf("%d %d %d ", a, b, c);
d = b-- || a++;
printf("%d %d %d ", a, b, d);
```

- a. 1 0 1 0 1
- b. 0 0 2 -1 1
- c. 1 1 2 1 1
- d. None

1.6 What is the output of the following code

```
#include <stdio.h>
void Interesting_Question() {
    int a[] = {1, 2, 3, 4, 5};
    int sum = 0;
    for(int i = 0; i < 5; i++) {
        if(i % 2 == 0) {
            sum += *(a + i);
        }
        else {
            sum -= *(a + i);
        }
    }
    printf("%d", sum);
}
int main() {
    Interesting_Question();
    return 0;
}
```

- a. 3
- b. 15
- c. -3
- d. None

1.7 What is the output of the following code?

```
#include <stdio.h>
void Solve_Karo() {
    int a = 10, b = 20, c;
    int c = a + b;
    {
        int c = b - a;
        printf("%d ", c);
    }
    printf("%d", c);
}
int main() {
    Solve_Karo();
    return 0;
}
```

- a. 10 30
- b. 10 10
- c. -10 30

d. None

What is the output of the following code?

```
#include <stdio.h>
```

```
int main() {
```

```
    printf("%d %d ", (023), (23));
```

```
    printf("%d %d %d", (076), (28), (0x87));
```

```
    return 0;
```

```
}
```

a. 23 23 76 28 0x87

b. 19 23 62 28 135

c. 023 23 076 28 135

d. None

What is the output of the following code?

```
#include <stdio.h>
```

```
union School {
```

```
    int age, rollNo;
```

```
    double marks;
```

```
};
```

```
int main()
```

```
{
```

```
    union School sc;
```

```
    sc.age = 19;
```

```
    sc.rollNo = 82;
```

```
    sc.marks = 19.04;
```

```
    printf("%d", (int)sizeof(sc));
```

```
    return 0;
```

```
}
```

a. 8

b. 16

c. 12

d. None

1.10 If p is an integer pointer with a value of 1000, then what will the value of p + 5 be?

a. 1020

b. 1040

c. 1010

d. 1005

Solve any Two of the following [5 Marks Each]

10M

a. Write a program to declare a one-dimensional array of size 50. Take inputs from the user in the array. Find the maximum element in the array and print it.

CO2, BL [3]

b. Write a C program to take input in two matrices of size 4x4 from the user. Perform matrix subtraction and store the result in another matrix of size 4x4, and print the subtraction of two matrices.

CO2, BL [3]

c. Implement the following string library function as a user-defined function 1. strcpy() 2. strcmp()

CO2, BL [3]

Solve any two of the following [5 Marks Each]

10M

a. Define function in C programming language. Also, explain the call by value and call by reference with examples.

CO3, BL [2]

b. Write a function to check if the given number is an Armstrong number or not, if the number is an Armstrong number then the function should return 1 otherwise 0. An Armstrong number, also known as a narcissistic number, is a number that is equal to the sum of its own digits, each

CO3, BL [3]

raised to the power of the number of digits. For example, 153 is an Armstrong number.

- c. Define Recursive Function. Write a recursive function to calculate the factorial of the given number.

CO3, BL [2]

Q4.

All questions are compulsory [5 Marks Each]

10M

- a. Define structure and union, and state the difference between them by using examples.
- b. In a class, the subject teacher wants to store the data of students on the computer. Therefore, it needs to develop the new data type which contains the following fields: Roll number (integer), name (character array), section (Character), year (integer) and marks (Fractional number). Create an array of size 50 and input all the details from the user and print it.

CO4, BL [2]

CO4, BL [3]

Q5.

Explain any two of the following function. [2.5 Marks Each]

5M

- a. malloc() b. calloc() c. free()

CO5, BL [2]